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The Impact of Investor Sentiment and Policy Uncertainty on Stock Market Volatility: Evidence from the Korean Stock Market

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ABSTRACT
This study forecasts the daily volatility of the Korean stock market using monthly investor sentiment and policy uncertainty through the GJR-GARCH-MIDAS model. Investor sentiment strongly forecast the future volatility. Among the various categorical policy uncertainty measures, economic policy uncertainty (EPU) provides the best volatility forecast. Subsample analyses suggest that the model with the realized variance and sentiment best forecasts the volatility, with EPU playing an important role in the post-COVID period.

KEYWORDS
GJR-GARCH-MIDAS; volatility; investor sentiment; policy uncertainty

JEL
C13; G17; G40

Introduction

Stock market volatility forecasting holds significant theoretical value and practical significance for both financial regulatory agencies and investors (Corsi 2009; S. Li and Liu 2025; Wang and Wu 2012). The GARCH-type (generalized autoregressive conditional heteroscedasticity) models have become a primary tool for volatility prediction. Among various GARCH-type models, the GJR-GARCH (Glosten-Jagannathan-Runkle GARCH) model captures the asymmetric volatility effect (volatility is negatively correlated with returns) and is widely applied in empirical finance (Conrad and Kleen 2020; Pan and Liu 2018). Another notable advancement is the Mixed Data Sampling (MIDAS) approach proposed by Ghysels, Santa-Clara, and Valkanov (2006), which allows for the integration of low-frequency and high-frequency variables. R. F. Engle, Ghysels, and Sohn (2013) extend this to the GARCH-MIDAS framework to incorporate macroeconomic information into volatility forecasts.

The Korean stock market, with a daily trading volume of approximately 64 billion dollars, is a leading emerging market in East Asia. Its derivatives market is highly active, ranking fifth globally in terms of trading volume by number of contracts in 2024. Although the volatility index from the options market (VKOSPI) exists, its informational value is constrained by microstructural limitations. Therefore, this study examines whether low-frequency predictors, such as investor sentiment and policy uncertainty, provide useful information for forecasting daily future volatility in the Korean market.

A substantial body of literature suggests that noise trader sentiment may lead to the overestimation or underestimation of fundamental information, thereby persistently affecting asset returns and resulting in irrational market outcomes (M. Baker and Wurgler 2006, 2007; Chung, Hung, and Yeh 2012). Additionally, drastic fluctuations in sentiment are often accompanied by heightened market volatility (Shen, Yu, and Zhao 2017), with such effects being particularly pronounced under extreme market conditions (Kumar and Lee 2006). Jiang and Jin (2021) find that investor sentiment exerts a significant positive influence on stock return volatility.

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Existing studies generally confirm that rising economic policy uncertainty (EPU) significantly suppresses stock market returns (Chen, Jiang, and Tong 2017; Guo, Zhu, and You 2018; Phan, Sharma, and Tran 2018). This phenomenon can be attributed to the fact that heightened policy uncertainty leads market participants to adopt a pessimistic outlook on the future economic environment, thereby reducing investment or adjusting their portfolios, which exerts downward pressure on stock prices. In terms of stock market volatility, research indicates that an increase in EPU exacerbates stock market fluctuations (T. Li et al. 2020; X. Liu and Ma 2025; Xue and Wang 2023; Yu, Fang, and Sun 2018). This effect may be due to the fact that rising policy uncertainty increases market information asymmetry, raises investors' required risk premiums, and reduces market liquidity, ultimately leading to greater stock price volatility.

Although extensive theoretical and empirical studies have examined the impact of investor sentiment and policy uncertainty on stock market volatility, few have incorporated both factors simultaneously into a volatility forecasting framework. To address this gap, this study employs the GJR-GARCH-MIDAS model in predicting the Korean stock market volatility and makes four key contributions. First, it jointly utilizes investor sentiment and policy uncertainty indicators to forecast long-run volatility. The former reflects investors' psychological expectations and biases, while the latter represents institutional risks in the macroeconomic environment. Second, this study disaggregates policy uncertainty into five categories—economic (EPU), fiscal (FPU), monetary (MPU), trade (TPU), and foreign exchange (FEPU)—and systematically examines their differentiated contributions to volatility forecasting. Third, it examines the role of the leverage effects by incorporating the GJR term. Finally, by analyzing the effects of low-frequency sentiment and policy uncertainty under different market regimes—including the 2008 global financial crisis, the 2020 COVID-19 high-volatility period, and the intervening low-volatility phase—this study provides novel empirical evidence for the Korean stock market and validates the state-dependent effectiveness of these predictors.

The main findings of this study can be summarized as follows. First, investor sentiment and EPU have a significantly positive impact on stock market volatility, with FPU showing marginal significance. Realized volatility and the asymmetric GJR term remain crucial predictors. Second, the out-of-sample tests indicate that the model achieves best performance when incorporating monthly realized volatility, investor sentiment, and EPU simultaneously. Third, subsample analyses reveal that investor sentiment consistently influences volatility across all periods, while EPU's impact is pronounced mainly during the high-volatility phase following COVID-19.

These results validate the effectiveness of the multivariate GJR-GARCH-MIDAS model for forecasting Korean stock market volatility using both sentiment and policy uncertainty measures. Contrary to prior research emphasizing the dominant role of sentiment (Birru and Young 2022), the evidence from this study suggests that sentiment and policy uncertainty are equally important when jointly considered. Among different categories of policy uncertainty, only FPU exhibits predictive power comparable to EPU, indicating their closer linkage to stock market fluctuations, with EPU being more robust overall. Investor sentiment appears slightly more influential, especially in low-volatility periods, whereas the predictive role of policy uncertainty strengthens during post-COVID high-volatility episodes.

Data

This study uses the Korean market data for stock volatility, sentiment, and policy uncertainty, over the period from January 2, 2006, to December 29, 2022,[1] determined by the availability of sentiment data. The daily KOSPI index level is obtained from the FnGuide database, and daily volatility is measured as r_i^2, where r_i denotes the log daily returns (in percentage terms). The monthly realized variance RV_t is computed as the squared sum of daily returns. Policy uncertainty data are sourced from Baker, Bloom, and Davis' website,[2] including five monthly indices—economic (EPU), fiscal (FPU), monetary (MPU),

trade (TPU), and foreign exchange (FEPU)—as developed by Cho and Kim (2022). The monthly percentage growth of the log indices is used.

Investor sentiment is constructed by combining variables proposed by Kim and Byun (2010) and Lee and Yoon (2018), followed by principal component analysis. The sentiment index ($SENT_t$) is defined as:

$$\begin{aligned} SENT_t = 0.227 * BSI_t + 0.564 * CD_t + 0.727 * TURN_{t-1} \\ -0.201 * CCSI_{t-1} - 0.248 * OPTION_{t-1} \end{aligned} \tag{1}$$

where *BSI* is individual investors' buy-sell imbalance, *CD* is the customers' deposits for stock investment, *TURN* is the turnover ratio, *CCSI* is the log consumer composite sentiment index by the Bank of Korea, and *OPTION* is the ratio of positive (call buy and put sell) minus negative (call sell and put buy) volume of the options market. For the analyses, the monthly percentage change in this sentiment variable is used. The first principal component explains 32.243% of the variance, capturing key aspects of investor sentiment.

The descriptive statistics are summarized in Table 1. Panel A presents the mean, standard deviation, and other univariate statistics. Panel B displays the correlation matrix. First, sentiment and uncertainty measures exhibit a slight positive correlation. Second, the correlations between sentiment and uncertainty measures are generally low, indicating that they move independently and capture different dimensions of information. Third, among the uncertainty measures, EPU and FPU show the strongest co-movement, with a correlation of 75%, while the other measures are less related to these two.

Estimation Model

The prediction model in this study is the GJR-GARCH-MIDAS model. It advances from the standard GARCH in two aspects. First, the GARCH-MIDAS model introduced by R. F. Engle, Ghysels, and Sohn (2013) decomposes the volatility into short-term and long-term components. Because it utilizes low-frequency predictors to forecast high-frequency volatility, the predictors are modeled to explain long-term components. The MIDAS component efficiently incorporates the long-term trends of low-frequency variables, while preserving high-frequency financial information, thereby accommodating mixed-frequency data. Moreover, the model is highly flexible and can incorporate multiple variables to examine the combined effects of different types of shocks. Second, GJR-GARCH specification captures the amplifying effect of negative shocks on a short-term volatility component, allowing for an effective representation of asymmetric market responses.

Equation (2) presents the GARCH-MIDAS specification of stock returns in day i of month t:

$$r_{i,t} = \mu + \sqrt{\tau_{i,t} * g_{i,t}}\varepsilon_{i,t} \ \forall i = 1, \ldots, N_t \tag{2}$$

$r_{i,t}$ represents the monthly log return; μ denotes the conditional expectation; τ_t stands for the long-term volatility component, $g_{i,t}$ represents the short-term volatility component; $\varepsilon_{i,t}$ is a standard Gaussian white noise sequence; N_t is the number of days in month t.

The GJR-GARCH(1,1) model (Conrad and Kleen 2020) is adopted for the short-term volatility $g_{i,t}$:

$$g_{i,t} = \left(1 - \alpha - \frac{\gamma}{2} - \beta\right) + \left(\alpha + \gamma * I_{\{\varepsilon_{i-1,t} < 0\}}\right) \frac{\varepsilon_{i-1,t}^2}{\tau_{i,t}} + \beta g_{i-1,t} \tag{3}$$

where $I_{\{\}}$ is an indicator function. The parameter γ (the GJR term) represents the presence of asymmetric volatility effects. The short-term component model must meet the criteria where $\alpha > 0; \beta \geq 0; \gamma \geq 0; \alpha + \beta + \frac{\gamma}{2} < 1$.

The long-term volatility $\tau_{i,t}$ is modeled using lagged exogenous variables $X_{j,i,t}$. Since the inclusion of an exogenous variable may have either a positive or negative effect, this study applies a logarithmic

Table 1. Summary statistics.

	Mean	Std.dev	Max	Min	skewness	kurtosis	J-B	ADF
Panel A: univariate statistics								
R	0.012	1.243	11.284	−11.774	−0.494	8.930	14,133.137***	−29.978***
RV	31.841	58.167	252.452	3.451	6.853	62.244	31,430.013***	−5.338***
ΔSENT	−0.774	63.673	169.155	−145.320	0.050	−0.300	0.847	−16.313***
ΔEPU	0.138	15.384	45.663	−60.644	−0.121	0.869	6.912***	−9.742***
ΔFPU	0.314	24.050	91.741	−64.477	0.472	0.906	14.541***	−9.624***
ΔFEPU	−0.027	46.500	134.934	−124.158	0.318	0.397	4.768*	−7.077***
ΔTPU	−0.172	46.421	182.687	−115.964	0.381	0.671	8.770**	−11.455***
ΔMPU	0.170	29.808	86.789	−79.800	0.132	−0.014	0.597	−13.953***
Panel B: Correlation matrix								
		RV	ΔSENT	ΔEPU	ΔFPU	ΔFEPU	ΔTPU	ΔMPU
RV		1.000						
ΔSENT		0.207	1.000					
ΔEPU		0.162	−0.134	1.000				
ΔFPU		0.165	−0.131	0.754	1.000			
ΔFEPU		0.149	−0.002	0.572	0.456	1.000		
ΔTPU		0.016	−0.086	0.249	0.131	0.143	1.000	
ΔMPU		0.194	−0.066	0.630	0.557	0.532	−0.018	1.000

Notes: Panel A shows the univariate summary statistics, and Panel B shows the correlation matrix. J-B stands for the Jarque-Bera test; ADF stands for the Augmented Dickey-Fuller unit root test results. ***, **, * represent significance at the 1%, 5%, and 10% levels, respectively. R is the daily stock return. RV is the monthly realized variance. SENT is the sentiment index. EPU, FPU, FEPU, TPU, MPU are categorical policy uncertainty about economic, fiscal, foreign exchange, trade, and monetary, respectively. Δ means the monthly difference, multiplied by 100.

transformation to the long-term volatility to ensure that $\tau_{i,t}$ remains positive. The multivariate specification is expressed as follows:

$$\ln \tau_{i,t} = m + \sum\nolimits_{j=1}^{L} \left(\theta_j \sum\nolimits_{k=1}^{K} \varphi_{j,k}(w_{j,1}, w_{j,2}) X_{j,i,t-k} \right) \tag{4}$$

where m is the constant term, X_j is the j-th exogenous variable, θ_j is the slope coefficient, and $\varphi_{j,k}$ is the beta weighting function.

$$\varphi_{j,k}(w_{j,1}, w_{j,2}) = \frac{\left(\frac{k}{K}\right)^{w_{j,1}-1}\left(1-\frac{k}{K}\right)^{w_{j,2}-1}}{\sum_{j=1}^{K}\left(\frac{j}{K}\right)^{w_{j,1}-1}\left(1-\frac{j}{K}\right)^{w_{j,2}-1}} \tag{5}$$

This study sets $w_{j,1} = 1$, ensuring a monotonically decreasing impact of each exogenous variable on long-term volatility. Consequently, the weight function is simplified as follows:

$$\varphi_{j,k}(w_2) = \frac{\left(1-\frac{k}{K}\right)^{w_{j,2}-1}}{\sum_{j=1}^{K}\left(1-\frac{j}{K}\right)^{w_{j,2}-1}} \tag{6}$$

The bandwidth parameter $K = 36$ is used following R. F. Engle, Ghysels, and Sohn (2013).

Equations (2), (3), (4), and (6) together defines the GJR-GARCH-MIDAS model. In Equation (3), the short-term component $g_{i,t}$ is modeled similarly to a GARCH(1,1) model, incorporating the GJR term γ for asymmetric volatility effect. In Equation (4), the log long-term component $\ln \tau_{i,t}$ is a linear combination of the exogenous variables. In a univariate case, $L = 1$. In a multivariate case, the exogenous variables can be considered as multiple regressors. The candidates of X_j are realized variance (RV), difference of sentiment ΔSENT, and difference of five policy uncertainty measures (ΔEPU, ΔFPU, ΔFEPU, ΔTPU, and ΔMPU).[3]

This study considers both the fixed window and rolling window approaches for determining the long-term component $\tau_{i,t}$ and low-frequency exogenous variables $X_{j,i,t}$. The fixed window uses fixed

values of $\tau_{i,t} = \tau_t$ and $X_{j,i,t} = X_{j,t}$ in a given month t. The rolling window allows for them to vary daily. If $X_j = RV$, $X_{j,i,t}$ is the sum of squared returns in previous 22 days. Otherwise, $X_{j,i,t}$ is a moving average of 22 days. For example, if $X_j = \Delta SENT$ and $i = 14$, then $X_{j,i,t} = \frac{1}{22}(8\Delta\text{SENT}_{t-1} + 14\Delta\text{SENT}_t)$.

Empirical Results

In-Sample Prediction

The parameter estimates for the single-factor GJR-GARCH-MIDAS model are reported in Table 2. The models without superscript means fixed window versions, and those with superscript (rw) are rolling window versions. The asymmetry parameter γ is always significantly positive, which supports the widely recognized presence of the leverage effect in equity markets.[4] This validates the extension to the GJR model, compared to the standard GARCH benchmark. The sum α + β + γ/2 is close to but less than one, suggesting strong persistence in short-term volatility. The rolling-window version of the GJR-GARCH-MIDAS model outperforms the fixed-window version both in log-likelihood (LLH) and BIC (Bayesian Information Criterion), indicating that the rolling-window specification more effectively captures time-varying volatility dynamics. Therefore, subsequent analyses focus solely on the results from the rolling-window version.

The MIDAS regression shows a significantly positive θ coefficient for realized volatility, indicating that monthly realized volatility positively affects the long-term volatility of stock returns. Investor sentiment also has a positive and significant effect, suggesting it improves volatility forecasting. Elevated sentiment often signals excessive optimism, stimulating speculative trading, lowering risk aversion, and strengthening herding behavior, thereby amplifying price fluctuations (Shu and Chang

Table 2. Parameter estimates of single-factor models.

Model	μ	α	β	γ	m	θ	w_2	*LLH*	*BIC*
RV	0.008	0.000	0.825***	0.201***	−0.520***	0.015***	6.814***	−4664.554	9386.154
	(0.015)	(0.006)	(0.016)	(0.019)	(0.060)	(0.001)	(0.853)		
$RV^{(rw)}$	0.008	0.000	0.809***	0.199***	−0.516***	0.019***	7.059***	−4627.983	9312.980
	(0.015)	(0.006)	(0.016)	(0.019)	(0.061)	(0.001)	(0.903)		
$\Delta SENT$	0.001	0.017**	0.886***	0.134***	0.061	0.025***	3.181***	−4693.974	9444.994
	(0.015)	(0.007)	(0.008)	(0.013)	(0.089)	(0.007)	(1.046)		
$\Delta SENT^{(rw)}$	0.004	0.015**	0.882***	0.140***	0.038	0.028***	3.605***	−4655.154	9367.323
	(0.015)	(0.007)	(0.008)	(0.012)	(0.081)	(0.006)	(0.910)		
ΔEPU	0.000	0.028***	0.879***	0.128***	0.061	0.145***	1.065***	−4700.215	9457.474
	(0.015)	(0.008)	(0.009)	(0.013)	(0.100)	(0.054)	(0.201)		
$\Delta EPU^{(rw)}$	0.002	0.022***	0.869***	0.142***	−0.046	0.254***	1.239***	−4660.631	9378.278
	(0.015)	(0.008)	(0.010)	(0.014)	(0.085)	(0.048)	(0.167)		
ΔFPU	0.008	0.030***	0.876***	0.129***	0.064	0.085**	1.230***	−4700.610	9458.266
	(0.015)	(0.007)	(0.009)	(0.013)	(0.100)	(0.034)	(0.294)		
$\Delta FPU^{(rw)}$	0.003	0.019***	0.845***	0.163***	−0.096	0.225***	1.389***	−4655.565	9368.145
	(0.015)	(0.007)	(0.012)	(0.016)	(0.071)	(0.028)	(0.147)		
$\Delta FEPU$	−0.001	0.029***	0.892***	0.119***	0.189	−0.004	12.234	−4700.660	9458.365
	(0.015)	(0.007)	(0.007)	(0.012)	(0.124)	(0.007)	(22.165)		
$\Delta FEPU^{(rw)}$	0.000	0.028***	0.885***	0.125***	0.137	0.027	1.192	−4665.054	9387.122
	(0.015)	(0.007)	(0.008)	(0.012)	(0.102)	(0.024)	(0.668)		
ΔTPU	−0.001	0.030***	0.893***	0.117***	0.218*	0.019	1.000	−4701.316	9459.676
	(0.015)	(0.007)	(0.007)	(0.012)	(0.130)	(0.019)	(0.462)		
$\Delta TPU^{(rw)}$	0.000	0.029***	0.888***	0.123***	0.172	0.000	1.000	−4665.341	9387.696
	(0.015)	(0.007)	(0.007)	(0.012)	(0.114)	(0.020)	(46.138)		
ΔMPU	−0.001	0.029***	0.892***	0.118***	0.192	−0.006	12.399	−4701.014	9459.073
	(0.015)	(0.007)	(0.007)	(0.012)	(0.123)	(0.012)	(27.923)		
$\Delta MPU^{(rw)}$	0.000	0.029***	0.889***	0.123***	0.183	−0.015	1.000	−4665.291	9387.597
	(0.015)	(0.007)	(0.008)	(0.011)	(0.116)	(0.037)	(1.300)		

Notes: This table reports the parameter estimates of all the single-factor models. LLH is the log-likelihood function and BIC indicates the Bayesian Information Criterion. The numbers in parentheses are the standard errors. ***, **, * represent significance at the 1%, 5%, and 10% levels, respectively. Models without superscripts are from fixed window approach, and those with superscript (rw) denote rolling window approach. The parameters are denoted in Equations (2), (3), (4), and (6).

2015). This aligns with noise trader-driven mispricing, where sentiment-induced distortions persist, sustaining elevated volatility.

Policy shocks can influence stock market volatility. During high volatility periods, media coverage on the market and economy tends to increase. However, some policy uncertainty measures may not relate directly to volatility, as certain policy expectations may already be priced in. Additionally, changes in market structure, other news, or international economic factors can weaken this relationship. Prior studies using categorized uncertainties report mixed findings; for example, L. Liu, Lü, and Yoon (2025) find that higher EPU and FPU increase market risk and volatility, while Xiao, Jiang, and Zhang (2024) show that only FPU positively affects both positive and negative volatility in the Chinese stock market.

The empirical findings show that the coefficients for θ on EPU and FPU are both positive and significant, indicating that policy uncertainty elevates the long-term volatility component of the stock market. This effect likely amplifies noise trading and investors' cognitive biases, thereby increasing market volatility. This result is consistent with R. Engle and Rangel (2008), who argue that macroeconomic uncertainty heightens market risk. As S. R. Baker, Bloom, and Davis (2016) highlight, economic and fiscal news represent the two most significant components of overall policy uncertainty. If future stock prices and volatility fully incorporate policy uncertainty, it explains why these two categories outperform others in predicting volatility and why they are closely linked to market fluctuations.

In contrast, θ coefficients for FEPU, TPU, and MPU are not statistically significant, suggesting these policy uncertainties have limited impact on long-term volatility. A possible reason is that FEPU and TPU reflect sector-specific uncertainties and less connected to the overall market. Despite Korea's trade dependency, uncertainties in foreign exchange and trade appear distant from aggregate market volatility. MPU also shows limited relevance, indicating that interest rate shocks in Korea are not closely tied to stock price movements.

Table 3 depicts the multivariate estimation results from rolling window versions. Those who are significant in univariate results in Table 2 are used. In the two-factor model incorporating investor sentiment and realized volatility, the θ coefficient for realized volatility decreases from 0.019 (single-factor) to 0.013, implying investor sentiment shifts market behavior toward emotion-driven decisions, increasing volatility and reducing realized volatility's explanatory power. The sentiment coefficient remains significantly positive, confirming its amplifying effect on long-term volatility. A similar pattern emerges after the inclusion of policy uncertainty. Among the policy measures, only the θ coefficient for EPU is significant at the 10% level, whereas FPU is not. Although EPU and FPU are highly correlated (75%), information contained in FPU may be largely subsumed by past volatility, while part of EPU captures information orthogonal to realized volatility. When both investor sentiment and policy uncertainty are included, the θ coefficient for sentiment decreases from 0.028 to 0.021 with EPU. EPU partially dampens investor confidence, leading to more cautious or withdrawal behavior. With fiscal policy uncertainty, the sentiment coefficient reduces further from 0.028 to 0.018, suggesting greater overlap between investor sentiment and fiscal policy effects on the market.

The three-factor model estimation indicates that all θ coefficients remain significant. Thus, investor sentiment, EPU, and FPU are all significant predictors of stock market volatility even after controlling for realized variance, a well-established predictor of future volatility. Despite some attenuation in individual coefficient magnitudes, each factor retains predictive power.

The enduring predictive power of investor sentiment, along with structural impacts of EPU and FPU, underscores information asymmetry in volatility adjustment. Rising uncertainty leads to rapid incorporation of negative information, while stabilizing effects emerge slowly due to bounded rationality and "expectation stickiness," prolonging elevated volatility (S. Li and Liu 2025). Overall, this study highlights, from behavioral finance and macro-finance perspectives, the critical roles of sentiment and policy uncertainty in shaping market risk dynamics. These insights deepen

Table 3. In-sample parameter estimates of multi-factor models.

Model	μ	α	β	γ	m	θ_1	θ_2	θ_3	$w_{2,1}$	$w_{2,2}$	$w_{2,3}$	*LLH*	*BIC*
RV+SENT	0.008	0.000	0.811***	0.195***	−0.487***	0.013***	0.019**		7.706***	1.000***		−4625.928	9325.161
	(0.015)	(0.008)	(0.016)	(0.019)	(0.063)	(0.001)	(0.008)		(1.116)	(0.208)			
RV+EPU	0.007	0.000	0.807***	0.197***	−0.526***	0.013***	0.075*		5.933***	2.221*		−4626.016	9325.337
	(0.015)	(0.007)	(0.017)	(0.019)	(0.062)	(0.001)	(0.045)		(0.922)	(1.265)			
RV+FPU	0.007	0.000	0.807***	0.199***	−0.494*	0.013***	0.049		7.354***	1.324***		−4627.207	9327.718
	(0.015)	(0.007)	(0.016)	(0.019)	(0.063)	(0.001)	(0.034)		(1.063)	(0.513)			
SENT+EPU	0.003	0.000	0.871***	0.163***	−0.087	0.021***	0.226***		4.540***	1.170***		−4650.787	9374.88
	(0.015)	(0.007)	(0.01)	(0.014)	(0.07)	(0.004)	(0.047)		(1.081)	(0.15)			
SENT+FPU	0.004	0.003	0.848***	0.175***	−0.116*	0.018***	0.201***		5.007***	1.294***		−4644.137	9361.579
	(0.015)	(0.007)	(0.013)	(0.017)	(0.064)	(0.004)	(0.028)		(1.243)	(0.134)			
RV+SENT+EPU	0.009	0.000	0.810***	0.193***	−0.556***	0.014***	0.004***	0.089**	4.160***	50.000**	2.448**	−4619.563	9328.721
	(0.015)	(0.009)	(0.017)	(0.019)	(0.066)	(0.001)	(0.001)	(0.044)	(0.757)	(22.753)	(1.17)		
RV+SENT+FPU	0.007	0.000	0.816***	0.190***	−0.531***	0.012***	0.017***	0.108***	1.000***	5.546***	1.978***	−4627.943	9345.481
	(0.015)	(0.007)	(0.017)	(0.019)	(0.019)	(0.002)	(0.003)	(0.043)	(0.207)	(1.402)	(0.57)		

Notes: This table reports the parameter estimates of all the rolling-window specification multi-factor models. The parameters θ_1, θ_2, and θ_3 ($w_{2,1}$, $w_{2,2}$, and $w_{2,3}$) are associated with the explanatory variables included in the model name, and they correspond to these variables in the exact order in which they are listed. LLH is the log-likelihood function and BIC indicates the Bayesian Information Criterion. The numbers in parentheses are the standard errors. ***, **, * represent significance at the 1%, 5%, and 10% levels, respectively.

understanding of volatility mechanisms and transmission across cycles, offering theoretical contributions and practical implications for market participants and policymakers.

Out-Of-Sample Prediction

This section focuses on the out-of-sample predictive ability of the model. In setting the prediction horizon, we initially set the final two years as the prediction period.[5] To rigorously assess the predictive performance of the models, the Model Confidence Set (MCS) methodology introduced by Hansen, Lunde, and Nason (2011) is employed. It evaluates forecasting errors through multiple loss functions. In light of the challenges emphasized by Diebold and Lopez (1996) regarding the selection of an optimal loss function for volatility prediction, three distinct criteria are used for comparison: Mean Squared Error (MSE), Mean Absolute Error (MAE), and the log-transformed R-squared (R^2 LOG). The GJR-GARCH(1,1) model is adopted as the benchmark, and the Diebold-Mariano (DM) test is conducted to statistically compare the out-of-sample forecasting accuracy of the extended models (Diebold and Mariano 2002), thereby determining the model with superior predictive capability.[6]

Table 4 reports the MCS test results for the GJR-GARCH-MIDAS model. Under the MSE loss function, all models pass the MCS test. Under the MAE and R^2 LOG loss functions, only the single-factor model with RV and the two three-factor models consistently pass the MCS test. Among the models considered, the model incorporating RV, SENT, and EPU demonstrates the best predictive performance according to the MCS evaluation. Table 5 reports the results from DM tests of the GJR-GARCH-MIDAS model. The models show no significant improvement under MSE, with some loss ratios above one. The model with RV, SENT, FPU reduces losses by 3.2% (MAE) and 2.5% (R^2 LOG). The model with RV, SENT, and EPU achieves the highest 3.5% loss reduction under MAE. Though results are similar when replacing EPU with FPU, FPU lacks robustness across model specifications (such as the out-of-sample forecast horizons). In unreported results, it is observed that the out-of-sample performance of the GJR-GARCH-MIDAS model is always better than the corresponding conventional GARCH-MIDAS model. Therefore, concerning the two-year-ahead forecasting capability, the model that simultaneously considers realized volatility, investor sentiment, and economic policy uncertainty demonstrates best predictive power regarding overall stock market volatility. Based on the success in the out-of-sample forecasts, we argue that both sentiment and EPU are important in forecasting volatility, even in more practical situations.

Table 4. The MCS test results for each model.

	MSE		MAE		R^2LOG	
MCS-2 years	T_R	T_{max}	T_R	T_{max}	T_R	T_{max}
GJR-GARCH(1,1)	**0.468**	**0.750**	0.027	**0.211**	**0.202**	**0.597**
RV	**0.406**	**0.951**	**1.000**	**1.000**	**0.880**	**0.843**
SENT	**0.066**	**0.912**	0.009	**0.221**	0.002	**0.459**
EPU	**0.468**	**0.912**	0.008	**0.221**	0.070	**0.459**
FPU	**0.406**	**0.620**	0.006	**0.221**	0.026	**0.459**
RV+SENT	**0.406**	**0.951**	**0.137**	**0.690**	0.070	**0.663**
RV+EPU	**0.679**	**0.951**	0.056	**0.762**	0.070	**0.597**
RV+FPU	**0.406**	**0.930**	0.027	**0.690**	0.014	**0.597**
SENT+EPU	**0.679**	**0.951**	0.002	**0.221**	0.000	**0.459**
SENT+FPU	**0.468**	**0.930**	0.009	**0.221**	0.008	**0.597**
RV+SENT+EPU	**1.000**	**1.000**	**0.916**	**0.916**	**0.961**	**0.961**
RV+SENT+FPU	**0.230**	**0.951**	**0.726**	**0.780**	**1.000**	**1.000**

Notes: This table reports the MCS p-values. Numbers that are larger than 0.10 are in bold and underlined, indicating that the corresponding model performs significantly better in the model set.

Table 5. Diebold-Mariano test results.

DM-2 years	MSE	MAE	R^2LOG
RV	0.995	0.965***	0.979***
	(−0.574)	(−4.950)	(−3.506)
SENT	0.999	1.002	1.003
	(−0.165)	(0.170)	(0.282)
EPU	0.997	0.996*	1.005
	(−0.845)	(−1.738)	(1.842)
FPU	1.001	0.992**	1.002
	(0.336)	(−2.556)	(0.689)
RV+SENT	0.993	0.972***	0.981**
	(−0.835)	(−3.472)	(−2.132)
RV+EPU	0.992	0.969***	0.988**
	(−0.906)	(−4.460)	(−2.130)
RV+FPU	0.996	0.972***	0.991
	(−0.551)	(−4.190)	(−1.666)
SENT+EPU	0.994	0.998	1.003
	(−0.662)	(−0.364)	(0.355)
SENT+FPU	0.997	0.992	0.994
	(−0.392)	(−1.144)	(−0.610)
RV+SENT+EPU	0.988	0.965***	0.975***
	(−1.462)	(−4.483)	(−3.106)
RV+SENT+FPU	0.993	0.968***	0.975***
	(−0.807)	(−3.927)	(−2.750)

Notes: This table shows the test results from the Diebold-Mariano (DM) test. The baseline model is the GJR-GARCH (1,1) model. T-statistics are in parentheses. ***, **, * represent significance at the 1%, 5%, and 10% levels, respectively.

Robustness Checks

One may argue that the main results are related to the volatility regime. Specifically, the out-of-sample period is affected by the COVID-19 pandemic and the subsequent high volatility market. Therefore, this section checks whether our results hold for various specification of sample period. The total sample from 2006 to 2022 includes two high-volatility periods and one low-volatility periods. The first high-volatility period is the global financial crisis period from 2006 to 2011. The second high-volatility period is the COVID-19 period from 2020 to 2022. The low-volatility period consists the rest, ranging from 2012 to 2019.

The summarized results from subsample analyses are presented in Table 6. The Benchmark row in Table 6 reports results from the full-sample analysis. Both sentiment and EPU are significant in-sample, and the model incorporating RV, SENT, EPU performs best out-of-sample model. In the Ex. COVID row, the COVID-19 period from 2020 to 2022 is excluded. It is observed that the significance of EPU weakens in-sample (becoming insignificant only in the two-factor model with RV). However, the three-factor model with RV, SENT, EPU is still the best out-of-sample model. In the Ex. GFC row, the global financial crisis period is excluded, focusing on the range from 2012 to 2022. In the EX. GFC and COVID row, the sample is further restricted to cover from 2012 to 2019. It is observed that the volatility prediction model weakens when the low-volatility period is used. Within this period, only SENT remains significant in-sample, while out-of-sample results are mixed, preventing identification of a robust predictive model across different fit criteria.

Since the GARCH model and its extensions are designed to capture volatility clustering during high volatility periods, the results in the low-volatility period are consistent with theoretical expectations. In conclusion, the sentiment index maintains strong predictive power across all subsamples, occasionally surpassing realized variance in importance. The role of EPU becomes more pronounced when the recent COVID-19 periods are included.

Table 6. Summary of subsample analyses.

Name	Time range	In-sample		Out-of-sample
Benchmark	2006 ~ 2022	RV	O	RV+SENT+EPU
		SENT	O	
		EPU	O	
Ex. COVID	2006 ~ 2019	RV	O	RV+SENT+EPU
		SENT	O	
		EPU	mixed	
Ex. GFC	2012 ~ 2022	RV	O	SENT
		SENT	O	
		EPU	mixed	
Ex. GFC and COVID	2012 ~ 2019	RV	X	mixed
		SENT	O	
		EPU	X	

Notes: This table shows the summary of the subsample analyses. The benchmark means the full sample. The Ex. COVID discards the COVID-19 period, starting from 2020. The Ex. GFC discards the global financial crisis period. The Ex. GFC and COVID ranges from 2012 to 2019. The out-of-sample period is set to two years.

Conclusion

This study investigates the effects of investor sentiment and policy uncertainty on daily volatility in the Korean stock market. The results indicate that investor sentiment significantly drives long-term volatility, with higher sentiment associated with increased volatility. Among various policy uncertainty indicators, only economic policy uncertainty (EPU) and financial policy uncertainty (FPU) show significant positive effects. Multivariate analysis confirms the joint influence of sentiment and these uncertainty measures. Realized volatility remains a key predictor, and the inclusion of the asymmetric GJR term enhances model performance. Out-of-sample forecasts reveal that incorporating realized volatility, sentiment, and EPU reduces prediction errors by approximately 3.5%. Subsample analysis shows that sentiment consistently predicts volatility across the full sample period, while the impact of EPU is particularly pronounced during the high-volatility phase following COVID-19.

The empirical results from the MIDAS model suggest that both the low-frequency sentiment and policy uncertainty are effective in forecasting high-frequency volatility after controlling for important factors (RV and the GJR term). Sentiment is related to price fluctuations from behavioral biases, and uncertainty is related to institutional risks. While they have different theoretical backgrounds, their joint effects on future volatility have been unclear. This study confirms that they have their own predictive ability, not subsumed by each other.

Policymakers should enhance the transparency and forward guidance of policy communication, establish a regular monitoring framework for sentiment and uncertainty indicators, and incorporate these measures into macroprudential assessments and stress-testing scenarios to enable early risk detection and timely intervention. In parallel, improving liquidity management and market stabilization mechanisms, increasing transaction transparency, and optimizing market structures can help mitigate the amplification effects of noise trading. Furthermore, strengthening investor protection and financial education can enhance market participants' ability to recognize and manage risks. Given that the conclusions of this study are subject to the constraints of the sample period and specific market conditions, these recommendations should be implemented cautiously, taking into account institutional contexts and pilot evaluations to reduce the impact of sentiment- and uncertainty-driven shocks while avoiding long-term market distortions.

This study focuses exclusively on the Korean stock market and does not cover other major markets such as the United States, China, or Japan, nor other leading emerging markets. Given differences and similarities in market mechanisms across countries, future research should investigate these dynamics individually. Additionally, using different sentiment and uncertainty indices—for example, Huang et al. (2015)'s Partial Least Squares (PLS) sentiment index to enhance its predictive power—would further emphasize the predictive role of sentiment and uncertainty.

Notes

1. A detailed description of the data construction is provided in Appendix A, available in the online supplementary material.
2. http://policyuncertainty.com/.
3. The input variables do not go through a principal component analysis (PCA) because it may change the meaning of the original variables. Since no mediators among them are assumed, no interaction terms are included. The variables of original scales are used (not normalized), and no anomalies are found from the scale of them.
4. In unreported results, it is observed that the models withγ performs better than those without γ in terms of log-likelihood and Bayesian Information Criterion.
5. Changing the prediction period to one or three years do not change our overall results.
6. A detailed description of the out-of-sample prediction test design is provided in Appendix B, available in the online supplementary material.

Disclosure Statement

No potential conflict of interest was reported by the author(s).

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